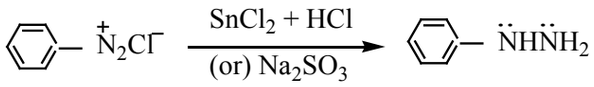
B. Reactions involving retention of diazo group

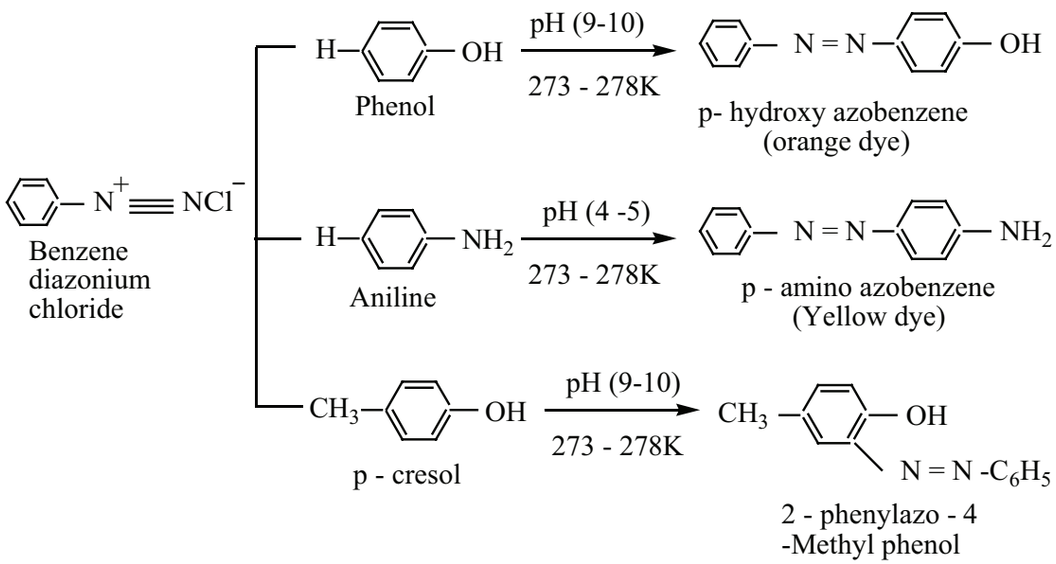
9. Reduction to hydrazines

Certain reducing agents like $SnCl_2$ / HCl; Zn dust / CH_3COOH, sodium hydrosulphite, sodium sulphite etc. reduce benzene diazonium chloride to phenyl hydrazine.



10. Coupling reactions

Benzene diazonium chloride reacts with electron rich aromatic compounds like phenol, aniline to form brightly coloured azo compounds. Coupling generally occurs at the para position. If para position is occupied then coupling occurs at the ortho position. Coupling tendency is enhanced if an electron donating group is present at the para – position to $-N_2^+Cl^-$ group. This is an electrophilic substitution.



Aryl fluorides and iodides cannot be prepared by direct halogenation and the cyano group cannot be introduced by nucleophilic substitution of chlorine in chlorbenzene. For introducing such a halide group, cyano group -OH, NO_2 etc.. benzenediazonium chloride is a very good intermediate Diazo compounds obtained from the coupling reactions of diazonium salts are coloured and are used as dyes.

13.4 CYANIDES AND ISOCYANIDES

13.4.1 Introduction

These are the derivatives of hydrocyanic acid (HCN), and is known to exist in two tautomeric forms



XII U13-Organic Nitrogen Compounds.indd 222 16-12-2022 19:45:26